3. Solvent Extraction.

Page No.
Date

1. Selection criteria of solvent for liquid - liquid Extraction.

Soln (1) Selectivity:

1) The ratio of the concn ratio of the solute to the feed solvent in the extract phase to that in the raffinate phase is called selectivity or seperation factor.
2) It is the measure of the effectiveness of solvent for seperating constituents of feed.
3) The selectivity should be greater than one for all useful extraction operation and if it is equal to one seperation by extraction is not possible.
4) If B is the solvent, feed contains A and C, where C is solute, and E and R are equilibrium phases, then selectivity β, of the solvent B is given by,

$$\beta = \frac{[\text{Wt fraction of } C / \text{Wt. fraction of } A]_E}{[\text{Wt fraction of } C / \text{Wt. fraction of } A]_R}$$

5) The selectivity has the same significance in Extraction as relative volatility in distillation.
6) The selectivity is the important factor in assessing the value of a solvent.
7) In extraction operation, the ease of seperation is directly related to the value of selectivity (β)
8). Poor selectivity means, large solvent to feed ratio and large number of extraction stages for a given degree of seperation.
9) β must be different from unity ($\beta > 1$), the higher the selectivity, the higher would be the seperation.

(2) distribution coefficient:

1) In dilute solutions at equilibrium, the ratio of the concentration of solute in two phases is called distribution coefficient.

2) It is given by,

$$K = \frac{C_E}{C_R}$$

where C_E and C_R are the concns of the solute in the extract and raffinate phases, respectively.

3) The distribution coefficient can also be given in terms of weight fraction of solute in the two phases in conta at equilibrium.

$$K' = \frac{y^*}{x}$$

where y^* is the wt fraction of the solute in extract and x is the wt fraction of solute in raffinate.

4) Higher values of distribution coefficient are generally desirable as the less amount of solvent and less no. of extraction stages are required for a given adsorption duty.

③ Recoverability:

1) As the solvent must be recovered for reuse usually by distillation, it should not form an azeotrope with the extracted solute.

2) For the low cost of recovery the relative volatility of the mixture formed should be high.

3) The latent heat of vaporisation of the solvent should be small whenever the solvent is to be vaporised.

(4) Capacity:

1) It is the ability of the solvent to dissolve the solute.

Page No.
Date

2) In ternary system with one Pair Partially miscible, solven can dissolve infinite amount of solute and this results in generally lower solvent requirements than for ternary syste m with two Pairs Partially miscible.

(5) Density:

The difference in densities of the saturated liquid phase should be larger for ease in the physical seperation of the phases.

(6) Insolubility of the solvent:

The solvent insoluble in the original liquid solvent should be Prefreered and it should have high solubility for the solute to be extracted as then Samall amount of solvents are required.

(7) Interfacial tension:

Interfacial tension should be high for the coalscence of emulsion to occur more readily as the same is of greater important than the dispersion.

(8) The solvent should be stable chemically and inert towards the other components and should not be corrosive towards the common material of construction.

(9). The solvent should be cheap.

(10) The solvent should be non-toxic and non flammable.

(11) The solvent should have low viscosity, freezing point, vapour pressure for ease in handling and storage.

Page No.
Date

Q. Differentiate between distillation and extraction.

Distillation	Extraction.
1) distillation is an operation in which constituents are seperated of liquid mixture are seperated using thermal energy.	1) Extraction is an operati in which liquid mixture ar seperated using insoluble liquid mixture.
2) distillation utilies the difference ~~betn~~ in Pressures and Volatility of different components at the same temp. to effect a seperation.	2) Extraction utilies the difference in solubility of different components to effect seperation.
3) In distillation, relative Volatility is used as a measure of the degree of seperation	3) In extraction, selectivity is used as a measure of degree of seperation.
4) In distillation, mixing and seperation of phases is easy and rapid	4) In extraction, the phases are hard to mix and harder to seperate.
5) distillation gives almost Pure Products.	5) Extraction itself does not g Pure Products and needs further Processing.
6) It does not offer more Flexi ~~Solubi~~lity in choice of operating conditions.	6) It does offer more flexibil in the choice of seperation
7) It requires thermal energy	7) It requires mechanical energy for mixing and sepera

$$\frac{(Y_1 - Y_S)}{(X_1 - X_F)} = \frac{-A}{B} \quad \text{--- (1)}$$

eqn (1) is the eqn of operating line for single stage operation.

7) The operating line passes through the pts. (X_F, Y_S) and (X_1, Y_1) and has a slope of $\frac{-A}{B}$

8) If the extraction solvent is pure then $Y_S = 0$ and so eqn (1) becomes

$$\frac{Y_1 - 0}{X_1 - X_F} = \frac{-A}{B}$$

$$\therefore \frac{Y_1}{X_1 - X_F} = \frac{-A}{B}$$

9) It is a straight line passes through pts $(X_F, 0)$ and (X_1, Y_1) and has a slope of $\frac{-A}{B}$

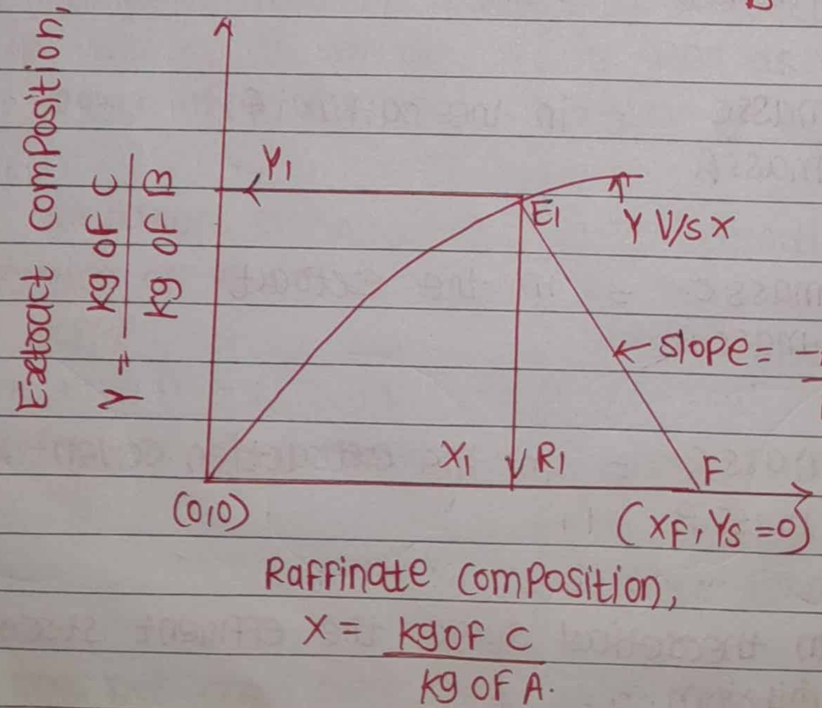


fig. single stage operation (insoluble solvents)

Q. Equations For counter current three stage solvent Extraction

Ans: Continuous multistage Countercurrent Extraction:

1) each circle represent the extraction arrangement involving a mixer and seperator.

2) Here, raffinate and extract streams Flow From stage to stage in countercurrent fashion.

3) In this type of arrangement, the Feed Solution (F) is Fed to the First stage which leaves as raffinate R_1, which then Passes through Further stages and leaves at n^{th} stage as Final raffinate.

4) The Fresh solvent enters the n^{th} stage and Passes through the stages in the opposite direction and Finally leaves the First stage as Finand Extract E_1,

5) so with this types of units we obtain Final Products (1) Raffinate (R_n) and (2) Extract (E_1)

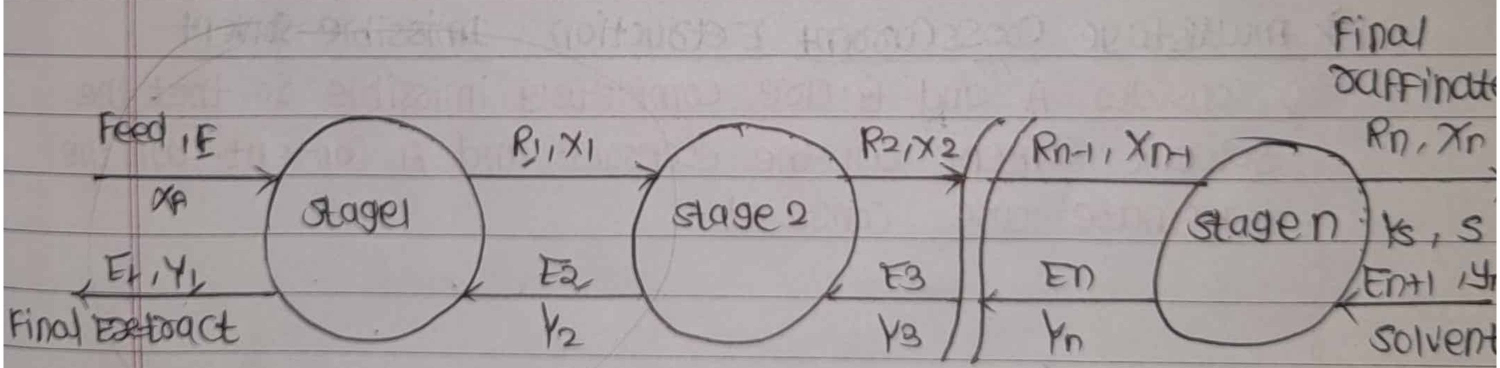


Fig. multistage counter current Extraction.

6) Solvent stream to the unit = S = E_{n+1} and $Y_S = Y_{n+1}$

material balance over stage-I

$\therefore F + E_2 = R_1 + E_1$

$\therefore F - E_1 = R_1 - E_2 = \Delta_R$

where Δ_R is the difference Point and it Represent new net Flow outward stage-I.

Q. Discuss on minimum solvent requirement for counter-current solvent Extraction in case of immisible solvents.

Ans:
1) Draw the equilibrium curve
2) locate pt L (x_n, Y_{n+1}).
3) Through X_F draw a vertical to cut the equilibrium curve at K.
4) Join KL
5) The slope of KL is equal to A/B min
6) with this slope find out B min i.e. min solvent required.

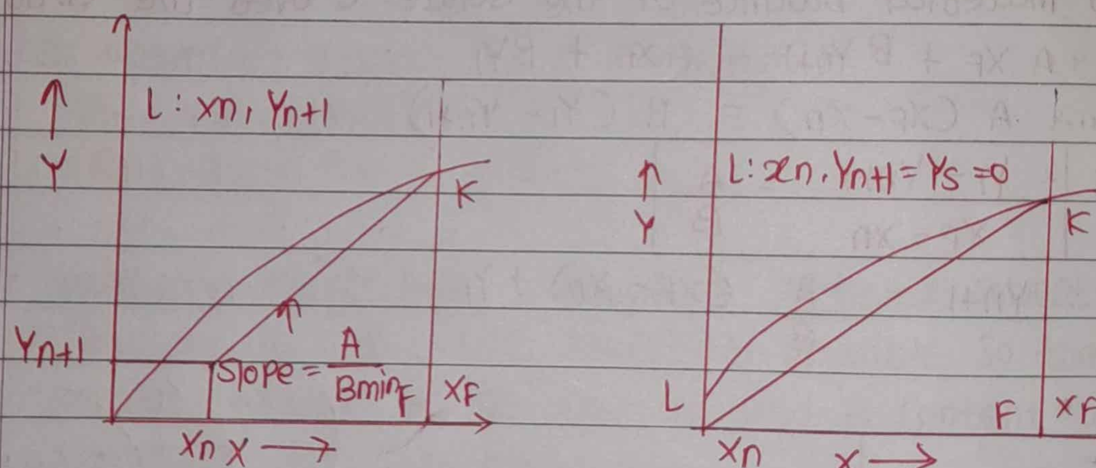
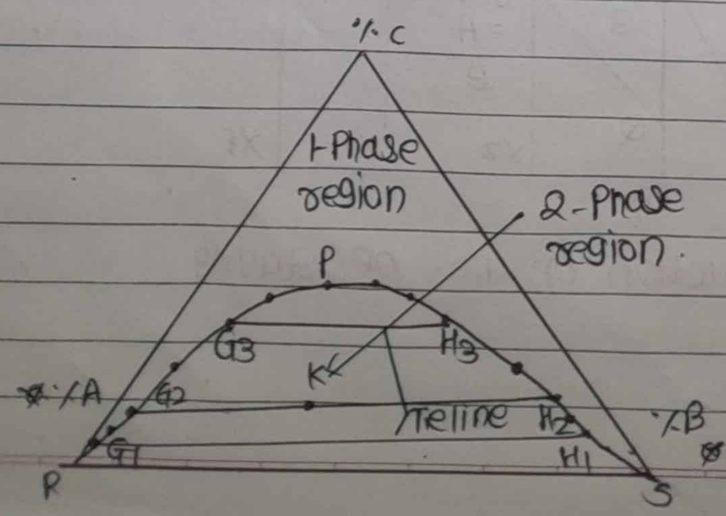


Fig. minimum solvent requirement (immiscible solvent)

Q. Define Binodal solubility curve. and Effect of temp. on Binodal solubility curve.

Ans:



1) Curve RPS is an equilibrium diagram known as Binodal solubility curve, because it has two arms RP and PS
2) G_1 and H_1 Two Phases in equilibrium.
H_1 is ~~rea~~ rich solvent and G_1 is raffinate.
3) $G_1 H_1$ = tie line
4) Point P = Plait Point
5) A liquid mixture having overall composition corresponding to Point K is a two Phase mixture.
6) on equilibrium it seperates two Phases of equilibrium
7) for tie line, G_2H_2 mixture rule gives the compositions as, $\frac{G_2}{H_2} = \frac{KH_2}{KG_2}$

Effect of temperature:

1) The effect of temperature on extraction operation can be shown with the three dimensional Figure where the temperature is Plotted vertically and the isothermal triangles are seen to be sections of the Prism
2) The mutual solubility of Feed solvent and extracting solvent increases with increasing the temp. of the mixture in the Extraction operation.
3) The increased solubility at higher temperatures influences the ternary equilibria considerly.
4) The critical sol^n temperature at which Phases dissolve completely to form homogeneous single phase which is not desiral
5) Not only does the area of heterogeneity decrease at high temp. but the slopes of the tie lines may also chane.